

Statistics – Module 3

Measures of Dispersion – Class 1

Exam Preparation Notes

What is Dispersion?

Dispersion tells us how much the observations in a dataset are spread out or scattered.

The topics covered in this class are:

1. Range
2. Coefficient of Range
3. Quartiles
4. Interquartile Range (IQR)
5. Quartile Deviation (QD)
6. Coefficient of Quartile Deviation
7. Quartiles in Individual Series
8. Quartiles in Discrete Series
9. Quartiles in Continuous Series

1 1. Range

Meaning

Range is the difference between the largest and smallest observations.

$$\text{Range} = L - S$$

where:

- L = largest value
- S = smallest value

Worked Question 1

Find the range:

12, 7, 19, 3, 15, 8

Largest value:

$$L = 19$$

Smallest value:

$$S = 3$$

Therefore:

$$R = 19 - 3 = \boxed{16}$$

Worked Question 2: Negative Values

Find the range:

-8, -3, 4, 11, -15, 7

Largest value:

$$L = 11$$

Smallest value:

$$S = -15$$

Therefore:

$$\begin{aligned} R &= 11 - (-15) \\ &= 11 + 15 \\ &= \boxed{26} \end{aligned}$$

Important

The data does not have to be sorted. Simply identify the maximum and minimum values.

1.1 Coefficient of Range

The coefficient of range expresses the range relative to the extreme values.

$$\boxed{\text{Coefficient of Range} = \frac{L - S}{L + S}}$$

Worked Question 3

For the data

$$10, 20, 30, 40, 50$$

find the range and coefficient of range.

Largest:

$$L = 50$$

Smallest:

$$S = 10$$

Range:

$$R = 50 - 10 = \boxed{40}$$

Coefficient of Range:

$$\frac{50 - 10}{50 + 10} = \frac{40}{60} = \frac{2}{3}$$

Therefore:

$$\boxed{\text{Coefficient of Range} \approx 0.667}$$

2. Quartiles

Quartiles divide ordered data into four parts.

$$Q_1 = 25^{\text{th}} \text{ percentile}$$

$$Q_2 = 50^{\text{th}} \text{ percentile} = \text{Median}$$

$$Q_3 = 75^{\text{th}} \text{ percentile}$$

Conceptually:

$$\text{Minimum} \quad | \quad Q_1 \quad | \quad Q_2 \quad | \quad Q_3 \quad | \quad \text{Maximum}$$

The four sections each contain approximately 25% of the observations.

Important Convention Used in Class

For individual/ungrouped data, the class uses:

$$\boxed{Q_1 = \text{size of } \frac{N+1}{4} \text{th item}}$$

$$Q_3 = \text{size of } \frac{3(N+1)}{4} \text{th item}$$

Here N is the number of observations.

3. Interquartile Range

The Interquartile Range measures the spread of the middle 50% of the data.

$$IQR = Q_3 - Q_1$$

It is the distance from Q_1 to Q_3 .

$$Q_1 \quad \underbrace{\hspace{2cm}}_{\text{middle 50\%}} \quad Q_3$$

Unlike ordinary range, IQR is less affected by extreme values.

4. Quartile Deviation

Quartile Deviation is also called the **Semi-Interquartile Range**.

It is half of the Interquartile Range.

$$QD = \frac{Q_3 - Q_1}{2}$$

Since:

$$IQR = Q_3 - Q_1$$

we can also write:

$$QD = \frac{IQR}{2}$$

5. Coefficient of Quartile Deviation

The coefficient of quartile deviation is a relative measure of quartile dispersion.

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

6. Individual Series – Worked Example

Question: From the following data, compute Interquartile Range, Quartile Deviation and Coefficient of Quartile Deviation.

24, 7, 11, 9, 17, 3, 20, 14, 4, 22, 27

Step 1: Arrange in ascending order

3, 4, 7, 9, 11, 14, 17, 20, 22, 24, 27

There are:

$$N = 11$$

Number the observations:

Position	1	2	3	4	5	6	7	8	9	10	11
X	3	4	7	9	11	14	17	20	22	24	27

Step 2: Find Q_1

$$Q_1 = \frac{N+1}{4} \text{th item}$$

$$= \frac{11+1}{4} = \frac{12}{4} = 3^{\text{rd}} \text{ item}$$

The 3rd item is 7.

$$Q_1 = 7$$

Step 3: Find Q_3

$$Q_3 = \frac{3(N+1)}{4} \text{th item}$$

$$= \frac{3(11+1)}{4} = \frac{36}{4} = 9^{\text{th}} \text{ item}$$

The 9th item is 22.

$$Q_3 = 22$$

Step 4: Interquartile Range

$$IQR = Q_3 - Q_1$$

$$= 22 - 7$$

$$\boxed{IQR = 15}$$

Step 5: Quartile Deviation

$$QD = \frac{Q_3 - Q_1}{2}$$

$$= \frac{22 - 7}{2} = \frac{15}{2}$$

$$\boxed{QD = 7.5}$$

Step 6: Coefficient of QD

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

$$= \frac{22 - 7}{22 + 7} = \frac{15}{29}$$

$$\boxed{\text{Coefficient of QD} \approx 0.52}$$

7 7. Discrete Series

What is Discrete Data?

In a discrete frequency distribution, we have:

- X = value
- f = frequency (how many times the value occurs)
- cf = cumulative frequency

Example:

X	f	cf
10	7	7
20	18	25
30	15	40
40	25	65
50	30	95
60	20	115
70	16	131
80	7	138
90	2	140

Meaning of Frequency

If:

$$X = 10, \quad f = 7$$

it means that the value 10 occurs 7 times.

Meaning of Cumulative Frequency

cf is the running total of frequencies.

For example:

$$7 + 18 = 25$$

$$25 + 15 = 40$$

Therefore, the first three cumulative frequencies are:

$$7, 25, 40$$

Why is cf useful?

The cumulative frequency tells us where each observation lies in the ordered dataset.

For the above table:

X	Observation positions
10	1–7
20	8–25
30	26–40
40	41–65
50	66–95
60	96–115
70	116–131
80	132–138
90	139–140

Key idea: The class/value tells us *what the observations are*; the frequency tells us *how many*; cumulative frequency tells us *how many observations have been covered so far*.

7.1 Worked Example: Discrete Series

Question: Calculate Quartile Deviation and its coefficient.

X	10	20	30	40	50	60	70	80	90
f	7	18	15	25	30	20	16	7	2

Step 1: Find N

$$N = \sum f$$

$$= 7 + 18 + 15 + 25 + 30 + 20 + 16 + 7 + 2$$

$$\boxed{N = 140}$$

Step 2: Construct cf

X	f	cf
10	7	7
20	18	25
30	15	40
40	25	65
50	30	95
60	20	115
70	16	131
80	7	138
90	2	140

Step 3: Find Q_1 position

Using the class method:

$$Q_1 = \frac{N+1}{4} \text{th item}$$

$$= \frac{140+1}{4} = \frac{141}{4} = 35.25^{\text{th}} \text{ item}$$

Now find the first cf that is equal to or greater than 35.25.

$$7, 25, \boxed{40}$$

The corresponding value of X is 30.

Therefore:

$$\boxed{Q_1 = 30}$$

Interpretation: the 35.25th observation has the value 30.

Step 4: Find Q_3 position

$$Q_3 = \frac{3(N+1)}{4} \text{th item}$$

$$= \frac{3(141)}{4} = \frac{423}{4} = 105.75^{\text{th}} \text{ item}$$

Find the first cf that is equal to or greater than 105.75:

$$7, 25, 40, 65, 95, \boxed{115}$$

The corresponding value is 60.

Therefore:

$$\boxed{Q_3 = 60}$$

Step 5: Quartile Deviation

$$QD = \frac{Q_3 - Q_1}{2}$$

$$= \frac{60 - 30}{2}$$

$$\boxed{QD = 15}$$

Step 6: Coefficient of QD

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

$$= \frac{60 - 30}{60 + 30} = \frac{30}{90}$$

Coefficient of QD = 0.33

Important Discrete-Series Trick

For discrete data:

Position $\rightarrow cf \rightarrow X$

That is:

1. Calculate the quartile position.
2. Look down the cf column.
3. Find the first cf that reaches or passes the position.
4. Take the corresponding X value.

8 8. Continuous Series**What is Continuous Data?**

In a continuous frequency distribution, values are grouped into class intervals.

Example:

Class interval	f
5–7	14
8–10	24
11–13	38
14–16	20
17–19	4

Here, f tells us how many observations belong to each class.

For example:

$$5-7, \quad f = 14$$

means there are 14 observations whose values lie in the 5–7 class.

It does **not** mean that observations numbered 5 through 7 are being referred to.

8.1 Worked Example: Continuous Series

Question: Calculate Quartile Deviation and its coefficient from:

Class interval	5–7	8–10	11–13	14–16	17–19
f	14	24	38	20	4

Step 1: Find N

$$N = 14 + 24 + 38 + 20 + 4$$

$$N = 100$$

Step 2: Construct cf

Class	f	cf
5–7	14	14
8–10	24	38
11–13	38	76
14–16	20	96
17–19	4	100

Step 3: Find the Q_1 class

For continuous series:

$$\frac{N}{4} = \frac{100}{4} = 25$$

We need the 25th observation.

Cumulative frequency:

Class	cf
5–7	14
8–10	38
11–13	76

The 25th observation falls between cf 14 and cf 38.

Therefore:

$$Q_1 \text{ class} = 8-10$$

Step 4: Continuous Q_1 Formula

$$Q_1 = L + \left(\frac{\frac{N}{4} - cf}{f} \right) i$$

where:

- L = lower boundary of the Q_1 class
- cf = cumulative frequency before the Q_1 class
- f = frequency of the Q_1 class
- i = class width

For the class 8–10:

$$L = 7.5, \quad cf = 14, \quad f = 24, \quad i = 3$$

Therefore:

$$\begin{aligned} Q_1 &= 7.5 + \left(\frac{25 - 14}{24} \right) 3 \\ &= 7.5 + \frac{11}{24} \times 3 \\ &= 7.5 + 1.375 \end{aligned}$$

$$Q_1 = 8.875$$

Note: The class slide used in class appears to show 8.75, but $7.5 + \frac{11}{24} \times 3$ gives 8.875. The displayed answer appears to contain an arithmetic/calculation inconsistency.

Step 5: Find the Q_3 class

$$\frac{3N}{4} = \frac{3(100)}{4} = 75$$

We need the 75th observation.

Cumulative frequencies:

$$38 < 75 \leq 76$$

Therefore:

$$Q_3 \text{ class} = 11\text{--}13$$

Step 6: Continuous Q_3 Formula

$$Q_3 = L + \left(\frac{\frac{3N}{4} - cf}{f} \right) i$$

For the class 11–13:

$$L = 10.5, \quad cf = 38, \quad f = 38, \quad i = 3$$

Therefore:

$$\begin{aligned} Q_3 &= 10.5 + \left(\frac{75 - 38}{38} \right) 3 \\ &= 10.5 + \frac{37}{38} \times 3 \\ &\approx 10.5 + 2.921 \end{aligned}$$

$$Q_3 \approx 13.42$$

8.2 Understanding Class Boundaries

This is important for continuous series.

The classes are:

$$5-7, \quad 8-10, \quad 11-13, \quad 14-16, \quad 17-19$$

There is a gap of 1 between consecutive class limits.

To obtain continuous boundaries, place the boundary halfway between them.

Between 10 and 11:

$$\frac{10 + 11}{2} = 10.5$$

Therefore:

$$11-13 \quad \longrightarrow \quad 10.5-13.5$$

Hence the lower boundary of 11–13 is:

$$L = 10.5$$

Similarly:

$$8-10 \longrightarrow 7.5-10.5$$

Thus:

$$\boxed{\text{Lower boundary} = \text{lower class limit} - 0.5}$$

for these integer-width classes.

Step 7: Quartile Deviation

Using the calculated values:

$$Q_1 = 8.875, \quad Q_3 \approx 13.42$$

$$\begin{aligned} QD &= \frac{Q_3 - Q_1}{2} \\ &= \frac{13.42 - 8.875}{2} \\ &\approx \boxed{2.27} \end{aligned}$$

Step 8: Coefficient of QD

$$\begin{aligned} \text{Coefficient of QD} &= \frac{Q_3 - Q_1}{Q_3 + Q_1} \\ &= \frac{13.42 - 8.875}{13.42 + 8.875} \\ &\approx \boxed{0.20} \end{aligned}$$

If following the displayed classroom value $Q_1 = 8.75$, the slide obtains approximately $QD = 2.34$ and coefficient 0.21. The discrepancy originates from the displayed Q_1 calculation.

9. Master Formula Sheet

Range

$$\boxed{R = L - S}$$

Coefficient of Range

$$\text{Coefficient of Range} = \frac{L - S}{L + S}$$

Individual Series: Q_1

$$Q_1 = \text{size of } \frac{N+1}{4} \text{th item}$$

Individual Series: Q_3

$$Q_3 = \text{size of } \frac{3(N+1)}{4} \text{th item}$$

Interquartile Range

$$IQR = Q_3 - Q_1$$

Quartile Deviation

$$QD = \frac{Q_3 - Q_1}{2}$$

Coefficient of Quartile Deviation

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Continuous Series: Q_1

$$Q_1 = L + \left(\frac{\frac{N}{4} - cf}{f} \right) i$$

Continuous Series: Q_3

$$Q_3 = L + \left(\frac{\frac{3N}{4} - cf}{f} \right) i$$

10 10. Exam Decision Guide

When you see a question, first identify the type of data.

Type	What to do
Individual data	Arrange values in ascending order and use the $(N + 1)$ positional formulas.
Discrete frequency data	Calculate N , construct cf, find the quartile position, then locate the corresponding X using cf.
Continuous frequency data	Calculate N , construct cf, identify the quartile class, then use the continuous-series quartile formula.

Discrete Data Shortcut

$$\boxed{\text{Position} \rightarrow cf \rightarrow X}$$

Continuous Data Shortcut

$$\boxed{\text{Position} \rightarrow \text{Quartile class} \rightarrow L, cf, f, i \rightarrow Q_1/Q_3}$$

11 11. Final Memory Sheet

1. **Range** = maximum minus minimum.
2. **Coefficient of Range** = $(L - S)/(L + S)$.
3. Q_1 represents the 25th percentile.
4. Q_2 is the median.
5. Q_3 represents the 75th percentile.
6. **IQR** = $Q_3 - Q_1$.
7. **QD** = $(Q_3 - Q_1)/2$.
8. **Coefficient of QD** = $(Q_3 - Q_1)/(Q_3 + Q_1)$.
9. In discrete data, **cf** tells you where an observation falls.
10. In continuous data, use the **quartile class** and the L, cf, f, i formula.
11. For the class intervals used here, the lower boundary is the lower limit minus 0.5.

12. Exam Workflow

For any quartile-dispersion problem:

Identify data type $\rightarrow$ Find $N \rightarrow$ Find $Q_1, Q_3 \rightarrow IQR \rightarrow QD \rightarrow$ Coefficient

Do not panic when you see a frequency table. The mathematics is mostly addition, locating a position, subtraction, division, and substitution into a formula.